## Hash function

k = 5 m = hashtable size.

1. Division Method → $K \% m$ $5 \% 10 = 5$
2. Midsquare Method → $K^2 = 3786 = 14333796$
3. Folding Method → 1347 → 13 + 47 = 60

### 10.2.2 Mid-square method

Here we have to take the binary representation of the key_value. Then we square these binary numbers and pick out the middle $k$ bits. Now these $k$ bits will be used as hash address. The size of the hash table will be (at least) $2^k - 1$.

$2^k$-1 is the Hash Table Size, if the Hash Function result exceeds it, we use reminder

Example: Binary form of *key_value* = 1011001, by squaring we get 1111011110001, let $k = 3$, then middle 3 bits = 111 = 7. So, the hash address is 7.

$K = \text{ceil}(\log_2(\text{SIZE}))$

**Mid-square Method**

In **mid-square method**, we first square the item, and then extract some portion of the resulting digits.

For example, if the item were 44, we would first compute 442=1,936. By extracting the middle two digits, 93, and performing the remainder step, we get 5 (93 % 11).

**Folding Method**

The **folding method** for constructing hash functions begins by dividing the item into equal-size pieces (the last piece may not be of equal size). These pieces are then added together to give the resulting hash value.

- For example, if our item was the phone number 436 – 555 - 4601, we would take the digits and divide them into groups of 2 (43, 65, 55, 46, 01) **(Fold Shifting)**.
- After the addition, 43 + 65 + 55 + 46 + 01, we get 210.
- If we assume our hash table has 11 slots, then we need to perform the extra step of dividing by 11 and keeping the remainder.
- In this case 210 % 11 is 1, so the phone number 436 – 555 - 4601 hashes to slot 1.